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UNOBTRUSIVE AUTHENTICATION SYSTEM

Abstract

Representative embodiments of performing a camera-based user authentication include determining whether a triggering event for authentication occurs; upon determining that the triggering event has occurred, causing a camera that is invisible to the user prior to occurrence of the triggering event to be visible to the user; causing the now-visible camera to acquire images associated with identifying information of the user; and based on the acquired identifying information, determining whether to authenticate the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/469,735 filed on Sep. 8, 2021, which claims priority to and the benefit of, and incorporates herein by reference in its entirety, U.S. Provisional Patent Application No. 63/078,590, filed on Sep. 15, 2020.

FIELD OF THE INVENTION

[0002] In various embodiments, the present invention relates generally to authentication and, more specifically, to a camera-based technique for user authentication.

BACKGROUND

[0003] Personal identification and security checks represent an integral part of maintaining physical and data security in a wide variety of environments, such as healthcare. Numerous systems and methods exist for identifying an individual based on personal biometric information. Unlike biometric techniques such as fingerprint recognition and the like, image-based authentication approaches such as facial recognition base identification on unique bio-information but do not require bodily contact with a recognition apparatus or even cooperation on the part of the individual seeking entry; such authentication systems are therefore versatile and popular. Conventionally, cameras may be deployed near physical entry points or adjacent secure workstations to acquire the identifying information from personnel seeking access. Such configurations, however, may intimidate some individuals and raise privacy concerns for others. Accordingly, there is a need for systems that perform image-based user authentication in an unobtrusive manner.

SUMMARY

[0004] Various embodiments hereof provide approaches to user authentication using a camera to capture the user's identifying information (such as facial features) but which remains concealed until needed, thereby eliminating (or at least reducing) anxiety and privacy concerns. For example, the camera, when unneeded for authentication, may be hidden behind a retractable shutter or “smart” barrier that can change its state from translucent to transparent and vice versa. In various embodiments, once a triggering event for authentication occurs (e.g., upon detecting the presence of a user or receiving a first authentication factor such as a password, a user identifier, an utterance for voice analysis, etc. from the user), the camera is made operable (and therefore temporarily visible to the user) for a prescribed time interval so as to allow the user to authenticate using the camera-based technique (e.g., face recognition). In addition, after expiration of the prescribed time interval or after the user successfully authenticates himself (or fails to authenticate himself over a predetermined number of attempts), the camera may be concealed again until the next triggering event for authentication occurs.

[0005] Accordingly, the present invention allows the camera in a security system to be concealed most of the time (e.g., when unneeded for authentication) and become visible only when acquiring the user's identifying information for authentication. This unobtrusive system avoids unnecessary intrusiveness and potential non-compliance with authentication protocols.

[0006] As used herein, the term “temporary” refers to a period of time adequate or necessary to complete a user authentication. In addition, reference throughout this specification to “one example,” “an example,” “one embodiment,” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the example is included in at least one example of the present technology. Thus, the occurrences of the phrases “in one example,” “in an

example,” “one embodiment,” or “an embodiment” in various places throughout this specification are not necessarily all referring to the same example. Furthermore, the particular features, structures, routines, steps, or characteristics may be combined in any suitable manner in one or more examples of the technology. The headings provided herein are for convenience only and are not intended to limit or interpret the scope or meaning of the claimed technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, with an emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

[0008] FIG. 1 schematically depicts an exemplary security system for performing user authentication in accordance with various embodiments of the present invention;

[0009] FIGS. 2A and 2B schematically depict exemplary shutters and a mechanism for concealing or exposing a camera in a security system in accordance with various embodiments of the present invention;

[0010] FIG. 2C schematically depicts components of an exemplary security system in accordance with various embodiments of the present invention;

[0011] FIG. 3 is a flow chart of an exemplary approach for performing a camera-based authentication in accordance with various embodiments of the present invention; and

[0012] FIG. 4 schematically illustrates a server for performing a camera-based user authentication in accordance with various embodiments of the present invention.

DETAILED DESCRIPTION

[0013] FIG. 1 depicts an exemplary security system **100** for authenticating a user **102** to permit access to a secure device (e.g., a computer or a medical device) **104**, a restricted application (which may be accessed via an otherwise unsecured device) and/or a secure region **106** in accordance herewith. In various embodiments, the security system **100** includes a camera **108** for acquiring identifying visual information of the user (e.g., facial features and/or an arbitrary physical object (“trinket”) chosen by or provided to the user), and a server **110** for processing the acquired images and, based thereon, authenticating the user. In addition, the security system **100** may include a reader **112** for reading and/or decoding identifying information provided by the user **102** for two-factor authentication. For example, the user may carry an identifying tag (e.g., a radio-frequency identification (RFID) tag, or a token with a bar code or a quick response (QR) code) or a mobile phone that can transmit the identifying information to the reader **112** via, for example, Wi-Fi LAN, Bluetooth wireless and/or near-field communications. Upon receiving and/or decoding the user-identifying information, the reader **112** may transmit this information to the server **110** to verify the user's identity. In one embodiment, the secure device **104** itself is capable of receiving identifying information from the user and transmitting the information to the server **110**. Alternatively, the security system **100** or the secure device **104** may include a user interface **114** for receiving, for example, a personal identification number (PIN), a password, or a fingerprint from the user for authentication.

[0014] In one embodiment, the security system **100** includes a presence-detection sensor **116** (e.g., an infrared sensor, a heat sensor, a proximity sensor, a pressure sensor, an acoustic sensor, or a motion sensor) for detecting the presence of the user at the secure device **104** and/or the entry point of the secure region **106**. The acoustic sensor may be, for example, a microphone, which detects user proximity by sound; the user may be required to utter a verbal command or statement, and

analysis of the sound clip may identify the user and serve as the first authentication factor.

Alternatively, the acoustic sensor may be sensitive to ultrasound frequencies and the user may be in possession of a wearable device that advertises its unique identifier using ultrasound over a limited spatial range; once again this modality can be used both to detect and identify the user as a first authentication factor.

[0015] The user's presence may alternatively be detected using RF, e.g., by wireless communication with the user's trinket or mobile device. In addition, the camera **108** may be normally concealed (e.g., substantially invisible to the user) until a triggering event occurs (e.g., the user arrives at or approaches the secure device **104** and/or the entry point of the secure region **106**). For example, following completion of user authentication and in the absence of a new triggering event, the camera **108** may be hidden behind a shutter or barrier **118**. In some embodiments, the orientation (e.g., angle and/or position) of the shutter **118** can be controlled by an adjustment mechanism **120** (e.g., a motor, a gimbal, or other manipulator that permits mechanical adjustment of the position of the shutter **118**). Upon detecting the presence or approach of a user, the sensor **116** may signal the adjustment mechanism **120** and/or the server **110** to change the orientation and/or position of the shutter **118** to reveal the camera **108** so that it may capture images of the user's identifying information (e.g., facial features or the trinket). For example, with reference to FIG. 2A, the adjustment mechanism **120** may physically rotate the shutter **118** around one or more axes so as to expose the lens of the camera **108** to the user. Alternatively, referring to FIG. 2B, the shutter **118** may include two panels **202**, **204**, and the adjustment mechanism **120** may open the panels **202**, **204** (e.g., along slide rails) to expose the camera **108**. In some embodiments, the adjustment mechanism **120** can further retract the panels **202**, **204** behind the camera **108**. After the camera **108** is revealed, the user may orient himself or the trinket to allow the camera **108** to obtain an image thereof for authentication.

[0016] Other arrangements not involving a shutter are possible. In one alternative, illustrated in FIG. 2C, the secure device **104** may have a conventional display screen **210** with which the user interacts via touch, voice, a keyboard, etc. A "pop-up" camera **215** is positioned behind the top of the display screen **210** and, when a user approaches the display screen **210**, the adjustment mechanism **120** extends the camera upward (as indicated by the arrow) so that it is now visible to, and can acquire an image of, the user. The pop-up camera **215** may be quite small for convenient mounting and operation, and to avoid visual prominence.

[0017] Alternatively, a small camera **215** may be recessed within the volume of the display screen **210**; when not in use, the top of the camera **215** is flush with (and preferably matches) the top edge **220** of the display screen **210**. The adjustment mechanism **120** may be a motor that bidirectionally drives the camera **215** along a vertical track, exposing and retracting the camera as appropriate. Similarly, the camera may be rotated into position; for example, the camera may be disposed in a cylindrical housing that is rotated from a first position, in which the wall of the cylinder (which may visually match the front of the secure device **104**) faces outward through a window in the secure device **104**, to an operative position with the camera lens positioned in the window. Still other alternatives involve folding the camera mount.

[0018] Additionally or alternatively, the triggering event may be determined by a security policy and involve events other than presence detection. For example, the event triggering uncovering of the camera may be a request by a secondary application (or an external website or server in communication with the secure device **104**) participating in multi-factor authentication of a user seeking access. This event may occur, for example, upon receipt of a first authentication factor from the user and determination that, for this user at this time, the security policy requires a secondary factor. In this case, the camera **108** may be concealed from the user until the user provides the first authentication factor to the reader **112**, secure device **104** and/or user interface **114** and the need for a biometric factor is determined. The first authentication factor may be, for example, information in a tag or mobile phone, a PIN, a password, or a fingerprint. In one

embodiment, upon receiving the first authentication factor, the receiving device (e.g., the reader **112**, secure device **104** and/or user interface **114**) determines whether biometric authentication is required and, if so, transmits a signal to the adjustment mechanism **120** to reveal the camera **108** and acquire images of the user and/or his trinket for authentication. The need for secondary authentication may also be based on analysis of a video feed from the vicinity of the secure device **104**, e.g., if a user switch or the presence of an additional user is detected.

[0019] Similarly, the event triggering concealment of the camera **108** can be determined by a policy, i.e., the event may be something other than completion of authentication. For example, the camera **108** may be hidden upon departure of the user (as detected by the camera **108** as the user moves out of view, or by a video feed from surveillance cameras).

[0020] It should be noted that the approaches described above for revealing the camera are exemplary only; any other suitable approaches for adjusting the relative orientations between the shutter **118** and the camera **108** so as to conceal and expose the camera **108** can be employed and are thus within the scope of the present invention. In addition, the first authentication factor may be received/decoded by any suitable device and may include any information identifying the user. Further, upon receiving the first authentication factor, the receiving device (e.g., the identifying information reader **112** or user interface **114**) may transmit a signal to the server **110**, which can then confirm the validity of the first authentication factor. If the first authentication factor is verified, the server **110** can signal the adjustment mechanism **120** to reveal the camera **108** as described above. Similarly, the adjustment mechanism **120** may be controlled by the server **110** in response to signals from the sensor **116**.

[0021] In other embodiments, the camera **108** is hidden behind a “smart” transparent barrier **118** whose light-transmission properties (e.g., opacity) can be altered by applying a voltage, light or heat thereto; in other words, the shutter need not be mechanical but can instead be optical. For example, the smart barrier may include or consist of an electrochromic material (e.g., tungsten oxide, WO₃) that is substantially opaque in the absence of a voltage; by “substantially opaque” is meant that the material is sufficiently untransmissive that a viewer cannot readily see the camera **108** through the activated material. In one embodiment, upon occurrence of the triggering event for authentication (e.g., detection of the user's presence and/or provision of the user's first authentication factor), the adjustment mechanism **120** may increase or decrease a voltage applied to the electrochromic material, thereby causing the smart barrier to switch its state from opaque to transparent. As a result, the camera **108** located behind the smart barrier can record an image of the user or the user's trinket for purposes of authentication. Alternatively, the smart barrier may include or consist of a micro-blind material having rolled thin metal blinds. With no voltage applied thereto, the metal blinds are rolled and allow light to pass; but upon application of a voltage, the rolled metal blinds stretch out and thus block light. In still another embodiment, the smart barrier includes or consists of a liquid crystal (LC, e.g., polymer-dispersed LC) shutter or light valve. Once again, increasing the applied voltage increases the material's opacity. Thus, in one embodiment, prior to occurrence of the triggering event for authentication, a voltage can be continuously applied to the micro-blind barrier or LC shutter to conceal the camera **108** therebehind. Upon occurrence of the triggering event, the circuit may be broken or the voltage reduced so as to make the barrier transparent.

[0022] It should be noted that the foregoing smart barrier technologies are exemplary only; other suitable materials (such as the photochromic material, thermochromic material, suspended particles, etc.) and their corresponding devices (e.g., a light source or a heat source) for switching the state of the barrier between substantially opaque and substantially transparent may be utilized and thus are within the scope of the present invention. In addition, the voltage, light, heat or other signal applied to the smart barrier may be controlled by the adjustment mechanism **120** and a voltage source, a light source or a heat source **122**. For example, upon detecting the user's presence using the sensor **116** and/or receiving the first authentication factor using the identifying

information reader **112** (and/or user interface **114**), the server **110** may transmit a signal to the voltage source (or light or heat source) **122** to apply or stop applying the voltage (or other signal) to the smart barrier, thereby changing its state. In some embodiments, the adjustment mechanism **120** includes a switch (a MEMS switch or a CMOS switch) for controlling the application of the voltage (or other signal) from the voltage source (or other signal source) **122** to the smart barrier in response to the triggering event.

[0023] In various embodiments, the camera **108** is visible to the user for a prescribed interval long enough to allow the user to authenticate herself. For example, if the camera-based authentication utilizes the user's facial features, the user may be given 10 seconds to orient her face to the camera **108**. In some embodiments, the security system **100** allows the user to choose the trinket for authentication. In this case, the user may be given a longer time (e.g., two minutes) to locate the trinket and present it to the camera **108**. Upon expiration of the prescribed time interval, the camera **108** may be concealed from the user again by, for example, changing the position of the shutter and/or the state of the smart barrier **118**. In some embodiments, after the camera **108** captures images of the user's identifying information (e.g., facial features or trinket), the camera **108** transmits the images to the server **110** for processing, and the server **110** thereupon attempts to verify the user's identity based on the transmitted images. If the user's identity is verified, the server **110** may grant access to the secure device **104** and/or secure region **106**. If, however, the identity cannot be verified, the server **100** may not allow the user to access the secure device **104** and/or secure region **106**.

[0024] In some embodiments, the camera **108** is visible to the user until the user successfully authenticates himself or fails to authenticate himself following a predetermined number of attempts (e.g., three). For example, upon verification by the server **110** of the user's identity based on the information provided by the camera **108**, the server **110** may grant access to the user and then cause the adjustment mechanism **120** to hide the camera **108** again. This spares the user from time pressure.

[0025] FIG. 3 illustrates an exemplary approach **300** for authenticating a user using a camera-based technique. In a first step **302**, the security system **100** is activated to detect a triggering event for authentication. For example, the sensor **116** may detect the presence of the user and the identifying information reader **112** and/or user interface **114** may receive the first authentication for the user. Prior to occurrence of the triggering event, the camera **108** utilized for performing the authentication is concealed behind the shutter or smart barrier **118**. In a second step **304**, upon detecting the triggering event, the adjustment mechanism **120** changes the orientation of the shutter and/or the state of the smart barrier **118** so as to allow the camera **108** to acquire images of the user's identifying information (e.g., facial features or a trinket). The camera **108** may be visible to the user for a prescribed time interval commensurate with the form of authentication. For example, the user may be accorded extra time to search for a trinket. In a third step **306**, the security system **100** determines whether the prescribed time interval has expired. If not, the position of the shutter and/or the state of the smart barrier remains unchanged for the user to authenticate himself (step **308**). If, however, the prescribed time interval has expired, the security system **100** causes the adjustment mechanism **120** to change the orientation of the shutter and/or the state of the smart barrier **118** so as to conceal the camera **108** from the user again (step **310**). In some embodiments, when the camera **108** is revealed, the security system **100** causes the camera to acquire images of the user's credentials (e.g., facial features and/or trinket) (step **312**), and based thereon determines whether to authenticate the user (step **314**). If the user has successfully authenticated himself, the security system **100** causes the adjustment mechanism **120** to change the orientation of the shutter and/or the state of the smart barrier **118** to conceal the camera **108** again (step **310**). Alternatively, if the security system **100** determines that the user has failed to authenticate himself after a predetermined number of attempts (e.g., three times) (step **316**), the security system **100** causes the camera **108** to be concealed from the user (step **310**). Otherwise, the security system **100** causes the

camera to remain visible (step 308) so that the user may again attempt to authenticate [0026] FIG. 4 illustrates an exemplary server 110 in accordance with embodiments of the present invention. The server 110 includes a controller 402 and system memory 404 including instructions, conceptually illustrated as a group of modules, that control the operation of controller 402 and its interaction with the other hardware components. The controller 402 executes commands and instructions and may be a general purpose computer, but may utilize any of a wide variety of other technologies including a special purpose computer, a microcomputer, minicomputer, mainframe computer, programmed microprocessor, micro-controller, peripheral integrated circuit element, a CSIC (customer-specific integrated circuit), ASIC (application-specific integrated circuit), a logic circuit, a digital signal processor, a programmable logic device, such as an FPGA (field-programmable gate array), PLD (programmable logic device), PLA (programmable logic array), RFID processor, smart chip, or any other device or arrangement of devices that is capable of implementing the approaches described above. In addition, the server 110 may include a communication component 406 (e.g., a modem, a network interface or other communication mechanism) designed to provide communications with a network—such as the Internet and/or any other land-based or wireless telecommunications network or system—and, through the network, with other devices (such as the secure device 104, identifying information reader 112, user interface 114, presence-detection sensor 116, adjustment mechanism 120 and/or voltage, light or heat source 122).

[0027] The system memory 404 includes a triggering-event determination module 408 for recognizing occurrence of triggering events based on information received from the sensor 116, secure device 104, identifying information reader 112 and/or user interface 114; an adjustment module 410 for controlling the adjustment mechanism 120 so as to change the position of the shutter 118 and/or the state of the smart barrier; an image-processing module 412 for processing the images acquired by the camera 108; an authentication module 414 for determining whether the user's credentials, provided via the images acquired by the camera 108, entitle her to access the secure device 104 and/or secure region 106; and a monitoring module 416 for monitoring the user's authentication progress (e.g., how much time the user has taken for authentication). The various modules may be programmed in any suitable programming language, including, without limitation, high-level languages such as C, C++, C#, Ada, Basic, Cobra, Fortran, Java, Lisp, Perl, Python, Ruby, or Object Pascal, or low-level assembly languages; in some embodiments, different modules are programmed in different languages. As will be readily understood by a person of skill in the art, the computational functionality required to carry out authentication methods in accordance herewith may be organized (in software modules or otherwise) in many different ways, and the depicted embodiment in FIG. 4 is, therefore, not to be regarded as limiting. Further details regarding suitable configurations of the server 110 are provided, for example, in U.S. Pat. Nos. 8,973,091 and 9,246,902, the entire disclosures of which are hereby incorporated by reference.

[0028] In general, the terms and expressions employed herein are used as terms and expressions of description and not of limitation, and there is no intention, in the use of such terms and expressions, of excluding any equivalents of the features shown and described or portions thereof. In addition, having described certain embodiments of the invention, it will be apparent to those of ordinary skill in the art that other embodiments incorporating the concepts disclosed herein may be used without departing from the spirit and scope of the invention. Accordingly, the described embodiments are to be considered in all respects as only illustrative and not restrictive.

Claims

1. A security system for a multi-factor authentication of a user, the system comprising: a camera for acquiring identifying information of a user, the camera being concealed from the user prior to occurrence of a triggering event for authentication; adjustment means for causing the camera to be

concealed from or revealed to the user; and a controller configured to: detect the triggering event and thereupon cause the adjustment means to reveal the camera to the user; acquire images of the identifying information of the user; and based on the acquired identifying information, determine whether to authenticate the user.

2. The security system of claim 1, wherein the adjustment means is mechanical.

3. The security system of claim 2, wherein the adjustment means comprises a shutter and an adjustment mechanism for controlling relative orientations of the shutter and the camera.

4. The security system of claim 1, wherein the adjustment means is optical.

5. The security system of claim 4, wherein the adjustment means comprises (i) a barrier having a selectable opacity level and comprising at least one of an electrochromic material, a photochromic material, a thermochromic material, suspended particles, polymer-dispersed liquid crystals or metal blinds and (ii) a signal source for setting the opacity level of the barrier.

6. The security system of claim 1, wherein the triggering event for authentication further comprises presence of the user.

7. The security system of claim 1, wherein the triggering event for authentication is determined by a security policy.

8. The security system of claim 6, further comprising a presence-detection sensor including at least one of an infrared sensor, a heat sensor, a proximity sensor, a pressure sensor, an acoustic sensor, an RF sensor or a motion sensor for detecting the presence of the user.

9. The security system of claim 1, further comprising at least one of an identifying information reader or a user interface for receiving a first authentication factor from the user.

10. The security system of claim 9, wherein the first authentication factor comprises at least one of a personal identification number, a token, a password, a bar code, a quick response code, or a fingerprint.

11. The security system of claim 1, wherein the controller is further configured to: determine whether a predetermined time interval has expired; and upon determining that the predetermined time interval has expired, cause the adjustment means to conceal the camera from the user.

12. The security system of claim 1, wherein the controller is further configured to: upon determining authentication of the user, cause the adjustment means to conceal the camera from the user.

13. The security system of claim 1, wherein the controller is further configured to: following a predetermined number of unsuccessful authentication attempts, cause the adjustment means to conceal the camera from the user.

14. A method of performing a multi-factor authentication of a user, the method comprising: detecting occurrence of a triggering event for authentication; revealing to the user a camera previously concealed from the user prior to occurrence of the triggering event; acquiring images associated with identifying information of the user; and based on the acquired identifying information, determining whether to authenticate the user.

15. The method of claim 14, wherein revealing the camera to the user comprises adjusting relative orientations of the camera and a shutter located in front the camera so as to cause the camera to be visible to the user.

16. The method of claim 14, wherein revealing the camera to the user comprises applying a signal to a variable-opacity element that (i) comprises at least one of an electrochromic material, a photochromic material, a thermochromic material, suspended particles, polymer-dispersed liquid crystals or metal blinds and (ii) is disposed in front of the camera so as to cause the camera to be visible to the user at a low opacity level.

17. The method of claim 14, wherein the triggering event for authentication further comprises presence of the user.

18. The method of claim 14, wherein the triggering event for authentication is determined by a security policy.

19. The method of claim 17, further comprising detecting the presence of the user with at least one of an infrared sensor, a heat sensor, a proximity sensor, a pressure sensor or a motion sensor.

20. The method of claim 14, further comprising: causing at least one of an information reader or a user interface to receive a first authentication factor from the user.
